

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – Industry Problem Solving Task

|                  |                                                                              |               |                                                   |
|------------------|------------------------------------------------------------------------------|---------------|---------------------------------------------------|
| Course Code:     | CSA06                                                                        | Course Title: | Design and Analysis of Algorithms                 |
| CO Assessed:     | CO4: Articulation (BL4, BL5)                                                 | Assessment:   | Assessment Tool 2 – Industry Problem Solving Task |
| Weightage:       | 20% of CIA                                                                   | Total Marks:  | 100 (2 Questions × 50 Marks)                      |
| CO4 Statement:   | Solve optimization problems using dynamic programming and greedy strategies. |               |                                                   |
| Student Name:    | Shaik Nuha                                                                   | Reg. No.:     | 192472242                                         |
| Section / Batch: | CSE-AI                                                                       | Date:         |                                                   |

### Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues.
- Thorough analysis with validated results and performance evaluation

| Question Type                 | Description                                                                            | Questions | Total Marks |
|-------------------------------|----------------------------------------------------------------------------------------|-----------|-------------|
| Industry Problem Solving Task | Focus on Solving optimization problems using dynamic programming and greedy strategies | Q1 – Q2   | 100         |

### SECTION A – Industry Problem Solving Task

#### Q103. Smart Factory Predictive Maintenance Scheduling (Time-Series + DP)

A manufacturing plant must schedule maintenance to prevent machine failures using sensor time-series data. Analyze how dynamic programming can optimize maintenance windows. Identify constraints such as production deadlines and maintenance crew availability. Design a model that balances downtime and failure risk. Discuss scalability across multiple production lines. Evaluate algorithm efficiency and prediction accuracy. Interpret results in reduced downtime and improved throughput.

#### Q104. Content Moderation Pipeline Optimization (Queueing + Priority Scheduling)

A social platform must process millions of content moderation requests with varying severity levels. Analyze how queueing models and priority scheduling can optimize processing. Identify constraints such as reviewer capacity and SLA deadlines. Design a system that dynamically prioritizes high-risk content. Discuss backpressure handling and burst traffic. Evaluate latency and throughput metrics. Interpret results in improved moderation effectiveness.

**Code of Conduct**

*I Shaik Nuha(192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_Sk.Nuha\_\_\_\_\_

**RUBRICS FOR CALCULATION****Industry Problem Solving Task**

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |
| Professionalism                     | 5%        | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |

**Mark Evaluation:**

| Case Study | Understanding of Industry Problem (20) | Application of Engineering Concepts (25) | Solution Design (25) | Use of Tools (15) | Analysis (10) | Professionalism (5) | Total Marks (Each - 50) |
|------------|----------------------------------------|------------------------------------------|----------------------|-------------------|---------------|---------------------|-------------------------|
| Q1         |                                        |                                          |                      |                   |               |                     |                         |
| Q2         |                                        |                                          |                      |                   |               |                     |                         |
|            | Overall Total (100)                    |                                          |                      |                   |               |                     |                         |

| Faculty In-charge               | Course Coordinator | Head of Department |
|---------------------------------|--------------------|--------------------|
| Name: _____<br>Signature: _____ | Signature: _____   | Signature: _____   |
| Date: _____                     | Date: _____        | Date: _____        |

**Q103.**

### **Aim**

To develop a predictive maintenance scheduling system using Dynamic Programming and time-series sensor data to minimize machine failures and production downtime.

### **Problem Statement**

A manufacturing plant must schedule maintenance for machines based on sensor data such as temperature and vibration. The goal is to determine the optimal maintenance schedule that minimizes downtime and failure risk while satisfying production deadlines and maintenance crew constraints.

### **Algorithm**

1. Collect sensor readings for each time period.
2. Predict failure risk for each period.
3. Initialize a DP array.
4. For each time period:
  - Calculate cost of performing maintenance.
  - Calculate cost of postponing maintenance.
  - Choose the minimum cost option.
5. Store the optimal decision in the DP table.
6. Output the minimum maintenance cost and schedule.

### **Python Code**

```
def predictive_maintenance(risk):
    n = len(risk)
    dp = [0] * (n + 1)
    for i in range(n - 1, -1, -1):
        maintain = 5 + dp[i + 1]    # maintenance cost
        skip = risk[i] + dp[i + 1]  # failure risk cost
```

```
dp[i] = min(maintain, skip)
return dp[0]
risk = [3, 8, 2, 6, 4]
print("Minimum Cost =", predictive_maintenance(risk))
```

### Sample Output

```
==== RESTART: C:\Users\nuhas\AppData\Local\Programs\Python\Python312\co4.py ====
Minimum Cost = 19
>>> |
```

### Conclusion

The Dynamic Programming approach helps determine optimal maintenance windows using predicted failure risks. This reduces unexpected breakdowns, minimizes downtime, and improves overall production efficiency.

### Q104.

#### Aim

To optimize content moderation processing using Queueing Models and Priority Scheduling for efficient handling of high-risk content.

#### Problem Statement

A social media platform receives millions of content moderation requests daily. The objective is to process requests efficiently while prioritizing high-severity content and meeting SLA deadlines.

#### Algorithm

1. Receive moderation requests with severity levels.
2. Assign a priority value to each request.
3. Insert requests into a priority queue.
4. Process highest-priority request first.
5. Continue until all requests are handled.
6. Measure throughput and response time.

## Python Code

```
import heapq

def moderation_system(tasks):
    pq = []
    for priority, task in tasks:
        heapq.heappush(pq, (-priority, task))
    print("Processing Order:")
    while pq:
        priority, task = heapq.heappop(pq)
        print(task)

tasks = [
    (5, "Violence Report"),
    (2, "Spam Content"),
    (4, "Hate Speech"),
    (1, "Minor Violation")
]

moderation_system(tasks)
```

```
>>> = RESTART: C:\Users\nuhas\AppData\Local\Programs\Python\Python312\co4.py
Processing Order:
Violence Report
Hate Speech
Spam Content
Minor Violation
>>> |
```

## Conclusion

Priority Scheduling combined with Queueing techniques ensures that high-risk content is reviewed first. This reduces moderation delays, improves platform safety, and increases overall system efficiency.